CM32024 / CM52037 Bayesian Machine Learning Coursework 2

Set by Dr Edward Milsom, Wednesday 18 March 2026.

Coursework 2 Overview

Part A: Sparse Variational Gaussian Processes. *Jupyter Notebook* **(35 Marks)**. Available on Moodle.

Part B (THIS DOCUMENT): Bayesian Optimisation on Preference Data. *PDF Report* **(15 Marks)**. Methodology from part A will be needed here.

Submission Instructions

Your submission should consist of two files:

- **A completed Jupyter notebook for part A:** All necessary instructions are contained in the part A notebook.
- **A PDF report for part B:** Instructions on what to include are given below.

In addition, you should submit a zip file containing all code used to generate the results and report for part B.

Part B: Bayesian Optimisation on Preference Data

Imagine you are a winemaker, and you want to find the most popular recipe for a new wine. Making new batches of wine with different ratios of ingredients is laborious and time consuming, but you can make a whole batch at once which is enough for many people to taste. Most people cannot give you a precise score of how much they like a particular wine, but given two different wines to try, they can tell you which one they prefer.

Hence you have devised a plan: you are going to use Bayesian optimisation on the *preference data* (binary choice between two wines) collected from groups of tasters to determine the best possible ratio of two ingredients for your wine. Through a series of rounds, you will make two batches of wine with different ratios of ingredients, gather preference data from multiple tasters, and use that data to update your surrogate model and select the next pair of wines to make. You will repeat this process until you have found the best possible recipe for your wine.

Bradley-Terry Likelihood for Preference Data

Given a latent function $f(\mathbf{x}) \in \mathbb{R}$ with $\mathbf{x} \in \mathbb{R}^2$, and pairs of inputs $\mathbf{x}_i, \mathbf{x}_j$, we assume the preference data are generated with the Bradley-Terry model:

$$p(y_{ij} = 1) = \sigma\left(f(\mathbf{x}_i) - f(\mathbf{x}_j)\right) = \frac{1}{1 + \exp\left(-(f(\mathbf{x}_i) - f(\mathbf{x}_j))\right)}$$

where $y_{ij} = 1$ indicates that wine $\mathbf{x}_i$ is preferred to wine $\mathbf{x}_j$, and $y_{ij} = 0$ indicates the opposite. Note that this means people's wine preferences can be noisy, i.e. there's a chance they prefer the wine which has a lower underlying "quality" (as given by the latent function f).

Note that the full data likelihood factorises over pairs of inputs:

$$p(\{y_{ij}\}|\{f(\mathbf{x}_i)\}) = \prod_{(i,j)} p(y_{ij}|f(\mathbf{x}_i), f(\mathbf{x}_j)) = \prod_{(i,j)} \sigma\left(f(\mathbf{x}_i) - f(\mathbf{x}_j)\right)^{y_{ij}} \left(1 - \sigma\left(f(\mathbf{x}_i) - f(\mathbf{x}_j)\right)\right)^{1-y_{ij}}.$$

Also note that, given $f(\mathbf{x}_i)$ and $f(\mathbf{x}_j)$ are jointly Gaussian distributed,

$$\begin{pmatrix} f(\mathbf{x}_i) \\ f(\mathbf{x}_j) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_i \\ \mu_j \end{pmatrix}, \begin{pmatrix} \Sigma_{ii} & \Sigma_{ij} \\ \Sigma_{ji} & \Sigma_{jj} \end{pmatrix} \right),$$

their difference $f(\mathbf{x}_i) - f(\mathbf{x}_j)$ is also Gaussian distributed:

$$f(\mathbf{x}_i) - f(\mathbf{x}_j) \sim \mathcal{N}\left(\mu_i - \mu_j, \Sigma_{ii} + \Sigma_{jj} - 2\Sigma_{ij}\right).$$

This will be important for computing the expected log likelihood correctly using a Monte Carlo estimator.

Thompson Sampling for Preference Data

The acquisitions functions we saw in the lecture cannot be directly applied to sample pairs of points in this setting (you will be asked to explain why in the discussion questions). Instead, we will use an alternative approach called *Thompson sampling*. Luckily, Thompson sampling is arguably simpler than using acquisition functions.

In Thompson sampling for regular Bayesian optimisation (i.e. when we have simple (x, y) observations instead of preference data), we sample a function from our surrogate model's posterior distribution (which represents a sample of what the true latent function might be), find the argmax of that sampled function, and take that argmax as the next input to gather data from. For our case of preference data, you will take 2 independent samples from the surrogate model posterior, find the argmax of each sample (by some kind of grid or random search), and take those two argmax points as the next pair of inputs to gather multiple preference data points between. Thompson sampling is simple yet inherently balances exploration and exploitation due to the randomness of the samples from the surrogate model posterior.

Latent Function and Preference Data Generation

You have been provided with code in `cw2_partB_data_generation.py` to generate preference data from a toy latent function. You *must* use the function `latent_function()` as your true latent function for this task, and you *must* use `generate_initial_preference_data()` to generate the initial preference data (do not change the default parameters, other than the seed to ensure the initial data isn't already on the global optimum). In `generate_initial_preference_data`, the variables `X_i` and `X_j` contain the first and second members, respectively, of each pair of points in the dataset, and `y_ij` contains the corresponding preference labels (therefore order is important!). When sampling new points during the Bayesian optimisation loop, you can use the `sample_preferences(rng_key, candidate_1, candidate_2, preference_samples_per_location=25):` function, which takes in 2 candidate points (i.e. a pair of wines made with different ingredients), and returns arrays `new_X_i`, `new_X_j`, and `new_y_ij` of preference samples to append to your dataset `X_i`, `X_j`, and `y_ij`. Note that `X_i` repeats `candidate_1` and `X_j` repeats `candidate_2` for each preference sample, so that they can be directly appended to the existing dataset of pairs and preferences.

Part B Report Requirements

You should implement a Bayesian optimisation loop using Thompson sampling, with a sparse GP surrogate function, to find the maximum of the above latent function using only preference data generated as in the above code. You should then produce a report, in `.pdf` format, which explains the methodology you used, and how you had to modify the code from part A, including screenshots or blocks of code, and plots of your resulting algorithm finding the global maximum of the latent function. In addition, you will be asked to answer some short questions at the end of your report. Further details on what to include in the report are given below, though you are welcome to include any additional information you think helps explain your approach and results.

Methodology and results (11 marks)

The 11 marks are broken down as follows:

- Surrogate GP for preference data: 5 marks
- Thompson Sampling: 3 marks
- Bayesian optimisation loop: 3 marks

To provide the necessary evidence for your work to be assessed, please write a single cohesive report on your approach to the surrogate model, Thompson sampling, and Bayesian optimisation procedure, covering the following points:

- Show and explain code blocks or screenshots of relevant changes to part A code. This should include:
  - The new likelihood function and computation of the expected log likelihood, taking care to only compute the elements of the kernel you actually need.
  - Implementing Thompson sampling to select pairs of points to gather preference data between.
  - Running the Bayesian optimisation loop, updating your surrogate model parameters each time you gather new preference data.
- Show the fit of your surrogate model on the *initial* preference data, **modifying the random seed to ensure the initial data are far away from the global optimum, so you can demonstrate that your algorithm finds the global optimum on its own**.
  - You should have 2 plots: one showing your surrogate model's mean prediction across the input space, and one showing the true latent function across the input space, with your collected data locations overlaid on both plots.
- Show the results of the Bayesian optimisation procedure after sampling at 50 new pairs (as chosen by your Thompson sampling procedure), with 25 preference samples per pair.
  - You should have 2 plots: one showing your surrogate model's mean prediction across the input space, and one showing the true latent function across the input space, with your collected data locations overlaid on both plots. It would be helpful to colour code the points in order of their collection, though this is not required for full marks.

Written questions (4 marks)

At the end of your report, please include a section giving short answers to the questions at the end of this document. Ensure you answer the correct questions for your unit code (CM32024 or CM52037).

Tips

- If you are encountering numerical issues and you're sure your equations are implemented correctly, try increasing the jitter you add to your kernel matrices to $1e-4$ or even $1e-3$ in extreme cases.
- Computing only the elements of the kernel required for the expected log likelihood is reasonably tricky, and will require a small amount of maths / creative thinking.
- Carefully consider when you need to sample from a joint distribution, or when you can get away with sampling from a marginal distribution (i.e. ignoring covariances).
- Don't forget to retrain your surrogate model after gathering new data!
- Since you won't have many datapoints to start with, it's probably easiest just to initialise the inducing points randomly.
- For convenience, you can use whatever batch size you like when training your surrogate model, and however many inducing points you like.

CM32024 Written Questions (4 marks)

Please note these questions are only for students on the CM32024 course. CM52037 students should answer the questions in the next section.

At the end of your report, please include a section giving short answers to the following questions:

1. Why does it make sense to use a **sparse** GP in this setting, rather than just a full-dataset variational inference approximation? (1 mark)
2. Why can we not (at least straightforwardly) use the acquisition functions we saw in the lectures for this task? (1 mark)
3. We could introduce a "temperature" parameter $\beta > 0$ to our likelihood function (the true one used to generate the data):

$$p(y_{ij} = 1) = \sigma\left(\frac{f(\mathbf{x}_i) - f(\mathbf{x}_j)}{\beta}\right) = \frac{1}{1 + \exp\left(-\frac{f(\mathbf{x}_i) - f(\mathbf{x}_j)}{\beta}\right)}.$$

   What effect does this temperature parameter have on the preference data we generate? (1 mark)
4. If $\beta >> 1$, what adjustment would be we need to make to our Bayesian optimisation loop to ensure we still get a well-fit surrogate model? (1 mark)

CM52037 Written Questions (4 marks)

Please note these questions are only for students on the CM52037 course. CM32024 students should answer the questions in the previous section. Also note that questions 1 in the CM52037 section consists of both questions 1 and 2 from CM32024, and question 2

consists of both questions 3 and 4 from CM32024, since CM52037 have slightly expanded learning objectives and must therefore do slightly more work.

At the end of your report, please include a section giving short answers to the following questions:

1. Why does it make sense to use a **sparse** GP in this setting specifically, rather than just a full-dataset variational inference approximation? **AND** why can we not (at least straightforwardly) use the acquisition functions we saw in the lectures for this task? (1 mark)
2. We could introduce a "temperature" parameter $\beta > 0$ to our likelihood function (the true one used to generate the data):

$$p(y_{ij} = 1) = \sigma\left(\frac{f(\mathbf{x}_i) - f(\mathbf{x}_j)}{\beta}\right) = \frac{1}{1 + \exp\left(-\frac{f(\mathbf{x}_i) - f(\mathbf{x}_j)}{\beta}\right)}.$$

   What effect does this temperature parameter have on the preference data we generate? **AND** if $\beta >> 1$, what adjustment would be we need to make to our Bayesian optimisation loop to ensure we still get a well-fit surrogate model? (1 mark)
3. Consider an alternative approach to this problem: we fit a surrogate model to the data using a small, cheap neural network (using some kind of Bradley-Terry loss to handle the preference data), or any other easy-to-use deterministic function approximator, and use an "ϵ-greedy" algorithm to select new points to evaluate, i.e. with probability ϵ we select a random point in the space, and with probability $1 - \epsilon$ we find the maximum of our neural network surrogate model's prediction and select that point. What are the strengths of the Bayesian optimisation approach we have taken compared to this alternative approach? (1 mark)
4. What are the weaknesses of the Bayesian optimisation approach we have taken compared to this alternative approach? (1 mark)